

Class #4 - Eigenvectors and Linear Transformations

1. Example: Let $B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$ and $D = \{\vec{d}_1, \vec{d}_2\}$ be bases for vector spaces V and W , respectively. Let $T : V \rightarrow W$ be a linear transformation with the property that T maps $\vec{b}_1$ to $3\vec{d}_1 - 5\vec{d}_2$, $\vec{b}_2$ to $-\vec{d}_1 + 6\vec{d}_2$, and $\vec{b}_3$ to $4\vec{d}_2$.

2. In general, let V and W be two finite dimensional vector spaces with dimensions n and m , respectively. Let $T : V \rightarrow W$ be any linear transformation. Let B and C be bases for V and W , respectively. Given any $x \in V$, the coordinate vector _____ is in _____. Then, the image $T(x)$ is in _____ and its coordinate vector _____ is in _____.

Hence, $[T(x)]_C = \text{_____} [x]_B$.

The matrix is a matrix representation of _____, and is called *the matrix for T relative to the bases _____ and _____*.

3. Example: Define $T : \mathbf{P}_3 \rightarrow \mathbf{R}^4$ by $T(p) = \begin{bmatrix} p(-3) \\ p(-1) \\ p(1) \\ p(3) \end{bmatrix}$.

(a) Show that T is a linear transformation.

(b) Find the matrix for T relative to the standard basis for $\mathbf{P}_3$ and the standard basis for $\mathbf{R}^4$.

4. Now suppose we have a finite dimensional vector space V with a basis B . If $T : V \rightarrow V$ is a linear transformation, then the matrix representation of T is called *the matrix for T relative to* _____, or simply the _____-matrix for T , and is denoted as _____.
5. Example: Let $B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$ be a basis for a vector space V . Find $T(3\vec{b}_1 - 4\vec{b}_2)$ when T is a linear transformation from V to V whose matrix relative to B is

$$[T]_B = \begin{bmatrix} 0 & -6 & 1 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix}.$$

6. Note that the matrix representation of T depends on the choice of basis vectors. Can we find a basis of V so that $[T]_B$ becomes a diagonal matrix?

Example: Define $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ by $T(x) = Ax$. Find a basis B for $\mathbf{R}^2$ with the property that $[T]_B$ is diagonal.

(a)

$$A = \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

...

(b)

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix}$$